

Differential Calculus — Implicit Differentiation

Worked Problem 01

Chapter 2: General Second-Degree Implicit Equation | Page 34

PROBLEM STATEMENT

Implicit Differentiation & Product Rule

Find $\frac{dy}{dx}$ from the given implicit algebraic relation:

$$ax^2 + 2hxy + by^2 + d = 0$$

STEP 1 প্রদত্ত দ্বিঘাত অব্যক্ত সমীকরণটি উল্লেখ করি (Given Implicit Second-Degree Equation)

$$ax^2 + 2hxy + by^2 + d = 0$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Differentiation Operator w.r.t. x)

$$\frac{d}{dx}(ax^2) + \frac{d}{dx}(2hxy) + \frac{d}{dx}(by^2) + \frac{d}{dx}(d) = 0$$

STEP 3 গুণন বিধি ও চেইন রুল প্রয়োগ করে প্রতিটি পদের অন্তরজ নির্ণয় করি (Applying Product Rule to xy & Chain Rule to y^2)

$$a(2x) + 2h\left(x\frac{dy}{dx} + y \cdot 1\right) + b\left(2y\frac{dy}{dx}\right) + 0 = 0$$
$$2ax + 2hx\frac{dy}{dx} + 2hy + 2by\frac{dy}{dx} = 0$$

STEP 4 উভয়পক্ষকে ২ দ্বারা ভাগ করে $\frac{dy}{dx}$ পদগুলো একপাশে সাজাই (Dividing by 2 & Grouping $\frac{dy}{dx}$ Terms)

$$(hx + by)\frac{dy}{dx} + (ax + hy) = 0 \implies (hx + by)\frac{dy}{dx} = -(ax + hy)$$

STEP 5 $\frac{dy}{dx}$ -এর জন্য সমাধান করে চূড়ান্ত অন্তরজ পাই (Solving for Explicit Derivative $\frac{dy}{dx}$)

$$\frac{dy}{dx} = -\frac{ax + hy}{hx + by}$$

FINAL ANSWER

$$\frac{dy}{dx} = -\frac{ax + hy}{hx + by}$$

■ BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- **Implicit Differentiation Principle:** Differentiating both sides with respect to x , treating y as an implicit function of x ($y = y(x)$).
- **Product Rule on Cross Term:** $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y$.
- **Chain Rule on Powers of y :** $\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$.
- **Constant Derivative:** $\frac{d}{dx}(d) = 0$.

⚠ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Treating y as a Constant:** Writing $\frac{d}{dx}(2hxy) = 2hy$ instead of using the product rule $2h(xy' + y)$.
- **Dropping $\frac{dy}{dx}$ on y^2 :** Writing $\frac{d}{dx}(by^2) = 2by$ instead of $2by\frac{dy}{dx}$.
- **Forgetting the Minus Sign on Transposition:** When moving $(ax + hy)$ to the RHS, it becomes $-(ax + hy)$.

💡 BOX 3: PARTIAL DERIVATIVE FORMULA SHORTCUT ($\frac{DY}{DX} = -\frac{F_X}{F_Y}$)

- For any implicit function $F(x, y) = ax^2 + 2hxy + by^2 + d = 0$:

$$F_x = \frac{\partial F}{\partial x} = 2ax + 2hy$$
$$F_y = \frac{\partial F}{\partial y} = 2hx + 2by$$
- Applying Euler's Implicit Function Theorem:

$$\frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}} = -\frac{2ax + 2hy}{2hx + 2by} = -\frac{ax + hy}{hx + by}$$
- This verifies the step-by-step solution instantly!

🌐 BOX 4: GEOMETRIC & CONIC SECTIONS INTERPRETATION

- **Conic Curve Tangents:** The equation $ax^2 + 2hxy + by^2 + d = 0$ represents a central conic (ellipse, hyperbola, or pair of lines).
- **Slope of Tangent:** At any point (x_1, y_1) on the curve, the slope is $m = -\frac{ax_1 + hy_1}{hx_1 + by_1}$.
- **Equation of Tangent:** Yields the standard conic tangent $axx_1 + h(xy_1 + yx_1) + byy_1 + d = 0$.

🎯 BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- **Step 1-2 (Implicit Differentiation Setup):** 1.0 Mark
- **Step 3 (Product Rule & Chain Rule Execution):** 1.5 Marks
- **Step 4 (Algebraic Grouping of $\frac{dy}{dx}$):** 0.5 Mark
- **Step 5 & Final Answer:** 1.0 Mark
- **Total Allocation:** 4.0 Marks